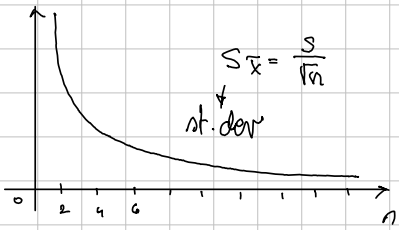
Nesigurnost tipa A – slučajna mjerna pogreška

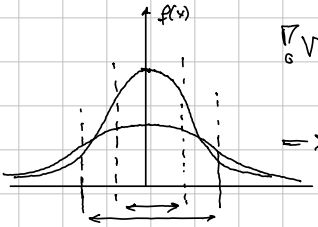
- nepredvidljivo, nasumično

$$u_A = \frac{a}{\sqrt{n}}$$

→ može se smanjiti ponavljanjem mjerenja veličine X ← mjerimo n puta

• standardno odstupanje od sredine
└ ponavljajuća mjerenja daju niz aritmetičkih sredina koje se mutisdno raseljavaju i raspadaju oko neke vrijednosti





veliki n nema smisla

⇒ Gaussova raspodjela, Studentova raspodjela za $n<30$

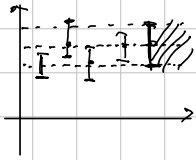
Nesigurnost tipa B – sistemni učinci

→ ostaju isti ili se mijenjaju na predvidljiv način

$$u_B = \frac{a}{\sqrt{3}}$$

• nesavršenost materijala, instrumenata ili postupka

$G = \pm a$



Složena nesigurnost C

$$\Rightarrow u_c = \sqrt{u_A^2 + u_B^2} \qquad x = \bar{x} \text{ uz } u_c(x)$$

$U_p = k_p \cdot u_c$ → $k_p = 2$ $p = 95\%$ $\quad x = \bar{x} \pm U_p$

$k_p = 3$ $p = 99\%$

---

① Zadane nesigurnosti napona i struje (u postotcima) kada je složena nesigurnost otpora? Npr 3% i 4%

$u_c(R) = ?$

Ohmov zakon: $R = \frac{U}{I}$

$$u_c(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial y}{\partial x_i}\right)^2 u^2(x_i)}$$

$$u_c(R) = \sqrt{\left(\frac{\partial R}{\partial U}\right)^2 \cdot u^2(U) + \left(\frac{\partial R}{\partial I}\right) \cdot u^2(I)} \rightarrow u_c(R) = R\sqrt{\left(\frac{u(U)}{U}\right)^2 + \left(\frac{u^2(I)}{I}\right)^2}$$

$$\frac{u_c(R)}{R} = \sqrt{\left(\frac{u(U)}{U}\right)^2 + \left(\frac{u(I)}{I}\right)^2}$$

ali ako je u postotcima → $\frac{u_c(R)}{R}$ je postotak!

$$\Rightarrow u(R\%) = \sqrt{u(U\%)^2 + u(I\%)^2} \Rightarrow u(R\%) = \sqrt{9+16} \rightarrow \boxed{u(R\%) = 5\%}$$

2. Ako znamo kombiniranu nesigurnost, koja je proširena nesigurnost uz mali broj mjerenja?

u_c – znamo

faktor proširenja $\left(\begin{array}{l} 95\% \Rightarrow 2 \\ 99\% \Rightarrow 3 \end{array}\right)$

$U_p = ? \rightarrow U_p = k_p \cdot u_c$

Efektivni stupanj slobode $k_p = n-1$

Za mali broj mjerenja (npr $n=5$) koristi se t raspodjela (studentova)

$$f(n,t) = K(n)\left[1 + \frac{t(n,P)}{n-1}\right]^{-\frac{n}{2}}$$

3. Ukupna relativna nesigurnost mjerenja manja ako je zadana rel. nesig napona i struje?

Relativna nesigurnost: $U_x = \frac{u(x)}{x}$, odnosno $U_x(\%) = \frac{u(x)}{x} \cdot 100\%$

$P = U \cdot I$, $u_p = \sqrt{\left(\frac{\partial P}{\partial U}\right)^2 \cdot u^2(U) + \left(\frac{\partial P}{\partial I}\right)^2 \cdot u^2(I)}$ $\quad U = \frac{I}{P} \quad I = \frac{P}{U}$

$$u_p = \sqrt{I^2 \cdot u^2(U) + U^2 \cdot u^2(I)} = P\sqrt{\left(\frac{u(U)}{U}\right)^2 + \left(\frac{u(I)}{I}\right)^2} \rightarrow \left(\frac{u_p}{P}\right) = \sqrt{U_U^2 + U_I^2}$$

$$U(P) = \sqrt{U_U^2 + U_I^2}$$

4. Izračunati neku složenu nesigurnost mjerene veličine $P = \frac{U^2}{R}$

$P = \frac{U^2}{R}$ $\quad u_c(P) = ? \rightarrow \sqrt{\sum_{i=1}^{n}\left(\frac{\partial P}{\partial x_i}\right)^2 u^2(x_i)}$

$\frac{\partial P}{\partial U} = 2\frac{U}{R}$ $\quad \frac{\partial P}{R} = U^2 \cdot \frac{-1}{R^2}$

$$u_c(P) = \sqrt{\left(2\frac{U}{R}\right)^2 \cdot u^2(U) + \left(-\frac{U^2}{R^2}\right)^2 u^2(R)} = \sqrt{\frac{U^4}{R^2}\left[\frac{4}{U^2}u^2(U) + \left(\frac{1}{R}u(R)\right)^2\right]}$$

$$u_c(P) = \frac{U^2}{R}\sqrt{4\left(\frac{u^2(U)}{U}\right)^2 + \left(\frac{u(R)}{R}\right)^2} \Rightarrow \frac{u_c(P)}{P} = \sqrt{4\left(\frac{u^2(U)}{U}\right)^2 + \left(\frac{u(R)}{R}\right)^2}$$